



BEFORE

Until the end of the 15th century, the New World and Old World had entirely separate agricultures, and most of the serious diseases of Europe and Asia had not crossed the Atlantic.

DOMESTICATED CROPS

New World domesticated crops—those whose growth was controlled by farmers—included **corn, tobacco, cocoa, and cotton**. The Old World was equally productive, but apart from cotton it shared few crop types with the Americas. Among its most important crops were **wheat, rice, and tea**.

DOMESTIC ANIMALS

Europe and Asia had an abundance of domesticated animals—such as **horses, cattle, goats, sheep, chickens, and pigs**—but in the pre-Columbian Americas the only equivalents were **turkeys, guinea pigs, dogs, alpacas, and llamas**. The



LLAMAS IN PERU

Americas, however, had an incredible range of wild animals, hence much meat in the diet was obtained through hunting and fishing.

DISEASE

Before Columbus, the Americas had enjoyed a long period of population growth. Nevertheless, **mortality rates remained high** from diseases such as tuberculosis, and waterborne parasites caused illness. The Europeans suffered similarly, but **they went on to export some of their deadly diseases**.

Columbus's very first voyage in 1492 (see pp.228–29) introduced new species to the Americas—he brought 28 horses, three mules, and an assortment of sheep, goats, cattle, chickens, dogs, and cats. Apart from the dogs, these were all unfamiliar creatures to the indigenous Americans, but Spanish colonization in the 16th century quickly expanded stock holdings. The numbers of domesticated animals imported by Europeans grew at different rates depending on the species, and brought different social effects. Sheep, goats, pigs, and chickens bred rapidly (the Europeans found excellent grazing land), and became useful not only as sustenance for the colonists, but also as foods to trade with native Americans for fruits and vegetables. Geographically, grazing animals also spread quickly—by 1519

sheep flocks could be found from northern South America up to the southwestern corner of what is today the United States.

Horses and cattle

Horse and cattle numbers grew more slowly because of the animals' longer breeding cycles and smaller numbers of offspring. Nevertheless, during the 16th century they spread through Peru and Chile and reached northward through New Spain, which covered Central America and much of southern North America. Horses were central to the early Spanish military conquests—many native Americans were at first terrified of horses, believing them to be divine creatures—but they also introduced effective transportation, providing the future means for a wider



Deadly trade

An early colonial artwork shows the lethal effect of smallpox on the native Americans. The first major smallpox outbreak in the Americas occurred between 1520–24, but epidemics returned regularly until the late 1800s.

The Great Exchange

Columbus's arrival in the Americas in 1492 began one of the greatest revolutions in global food habits ever seen. Plants and animals previously separated by some 3,000–4,000 miles (5,000–6,000 km) of ocean would now be exchanged.

